

Chapter 5:

Sexually Transmitted Diseases

By: Fantahun Guteta (BscN, Mph)

Introduction

- The diseases belonging to this group are usually transmitted during sexual intercourse; hence the name sexually transmitted diseases or STDs.
- During sexual intercourse there is close body contact, which is an ideal situation for transmission.
- The causative organisms of the STDs are very easily killed by drying or by cooling to below body temperature.
- ***Therefore*** transmission of these agents can only occur under very special circumstances, mostly during ***sexual intercourse.***

Introduction...cont'd

- STIs are serious and common problems worldwide.
- There are more than 20 types.
- Many of these are **curable** with effective treatment, but continue to be a major health problem for an individual and the community at large.

Risk factors

- **Age:** 15 years and older
- **Marital status:** unmarried people who often change their sexual partners
- **Occupation:** soldiers, policemen, students, seasonal laborers, and other people who are temporarily away from home tend to expose themselves more easily.
- **Residence:** urban area are more likely affected
- **Commercial sex workers**

1. Syphilis (Hard chancre)

Definition

- A disease characterized by a primary lesion, a later secondary eruption on the skin and mucus membranes, then a long period of latency, and finally late lesions of skin, bones, viscera, CNS and cardiovascular systems.

Infectious agent

- *Treponema pallidum*, a spirochete.

Syphilis...



Syphilis...

Epidemiology

Occurrence:

- Worldwide spread.
- Primarily involving sexually active young people between 20 and 29 years.
- More common in urban than rural areas.

Reservoir - Humans

Mode of transmission:-

- by direct contact with lesion mainly during sexual intercourse.
- Accidentally by touching infective tissues or via blood transfusion or congenitally, which may occur before birth, in the case of an infected mother.

Clinical Manifestation

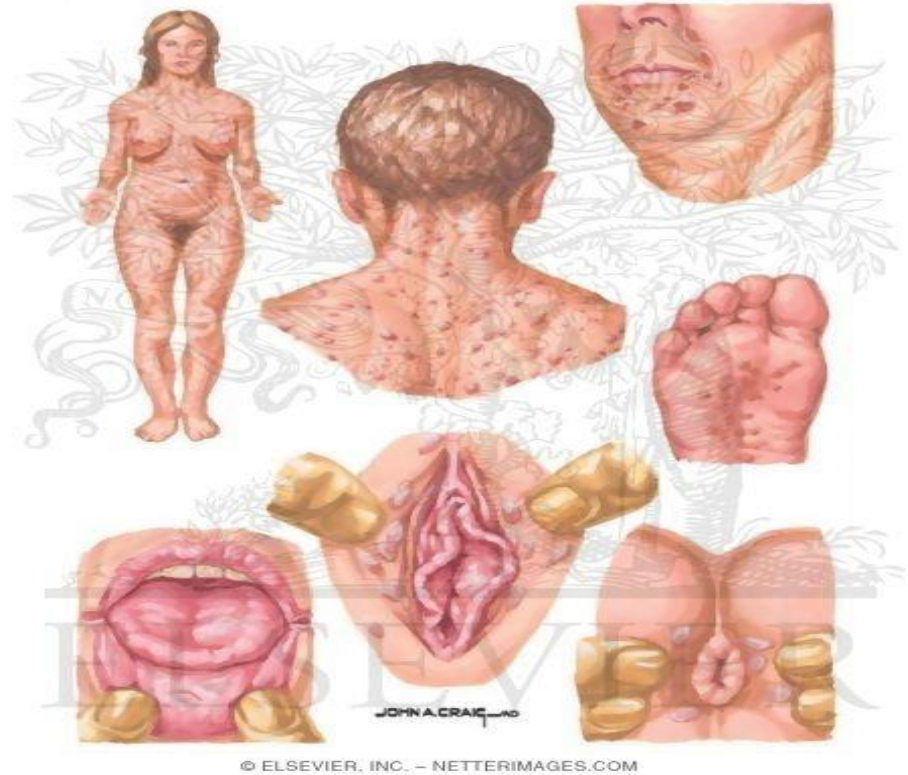
- The clinical presentation is divided into **three groups**:
- a) Primary syphilis** –
 - consists of hard chancre, the primary lesion of syphilis, together with regional lymphadenitis.
 - The hard chancre is a single, **painless ulcer** on the genitalia or elsewhere (lips, tongue, breasts) and heals spontaneously in a few weeks without treatment.
 - The lymph glands are bilaterally enlarged and not painful.

Primary syphilis...cont'd



b) Secondary syphilis –

- After 4 – 6 weeks of the primary infection, a generalized secondary eruption appears, Rashes, Skin lesions



c) Tertiary syphilis

- This stage is characterized by destructive, non-infectious lesions of the skin, bones, viscera, and mucosal surfaces.
- Other disabling manifestations occur in the cardiovascular system (aortic incompetence, aneurysms) or central nervous system (dementia paralytica, tabes dorsalis).



Syphilis in pregnancy

- According to the severity, congenital syphilis can result in **congenital abnormalities**, still birth, or repeated spontaneous abortions.



Syphilis...

Prevention and control

- Treatment of cases
- Treatment of contacts and source of infection
- Health education on safe sex
- Controlling STDs among commercial sex workers
 - ✓ Monthly check up and treatment of cases
 - ✓ Provision of condom
- Screening of pregnant women and early treatment to prevent congenital syphilis
- Screening of blood before transfusion.

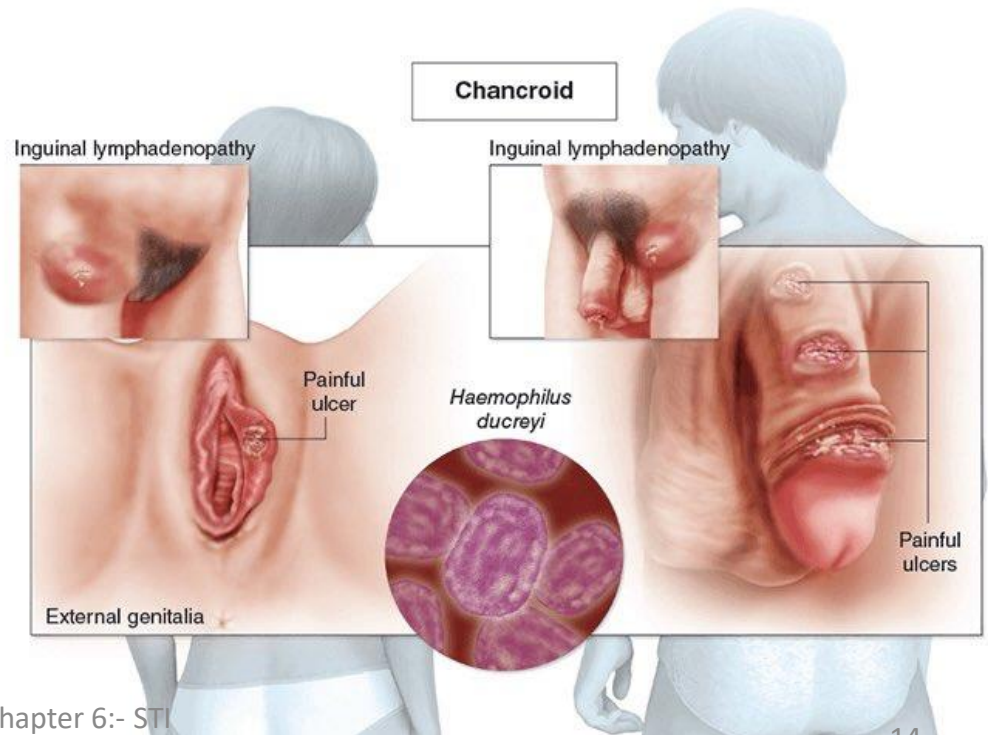
2. Chancroid (soft chancre)

Definition

- An acute bacterial infection localized in the genital area and characterized clinically by single or multiple painful necrotizing ulcers at the site of infection.

Infectious agent.

- *Haemophilus ducreyi*



Chancroid...cont'd

Epidemiology

Occurrence

- Endemic in many developing countries.
- The commonest cause of genital ulcer in many developing countries.

Reservoir – Humans

Mode of transmission

- by direct sexual contact with discharges from open lesion and pus from buboes.

Chancroid...

Clinical manifestation

- Classic Chancroid ulcer begins as a tender papule that ulcerates within 24 hours.
- The **ulcer is painful**, irregular and sharply demarcated from the nearby skin.



Prevention and control

- Case treatment
- Investigation of contacts, source of infection and treatment
- Thorough washing of genitalia with soap and water promptly after intercourse is very effective.
- Controlling STDs among commercial sex workers
- Sex education for high risk groups

3. Lymphogranuloma venereum

Definition

- A venereal disease caused by chlamydia microorganisms, most commonly manifested by acute inguinal lymphadenitis.

Infectious agent

- Chlamydia trachomatis



Lymphogranuloma venereum...

Epidemiology

Occurrence

- Common in most parts of the world.
- Its incidence is more common in males than females.

Reservoir

- Humans often asymptomatic (particularly in females)

Mode of transmission

- Direct contact with open lesions of infected people, usually during sexual intercourse.

Lymphogranuloma venereum...

Clinical manifestation

- Lymphadenopathy with non-specific symptoms of
 - ✓ fever,
 - ✓ chills,
 - ✓ headache,
 - ✓ malaise,
 - ✓ anorexia and
 - ✓ weight loss.

Lymphogranuloma venereum...

- In the female, inguinal nodes are less frequently affected and involvement is mainly of the pelvic nodes with extension to the rectum and recto vaginal septum, resulting in **proctitis**, stricture of the rectum and fistula.
- **Elephantiasis of genitalia**, scrotum and vulva occur in either sex.

Prevention and control

- Early diagnosis and treatment of cases
- Investigation of contacts, source of infection and treatment
- Control STDs among commercial sex workers
- Sex education for high risk groups

Herpes Genitalia

Definition

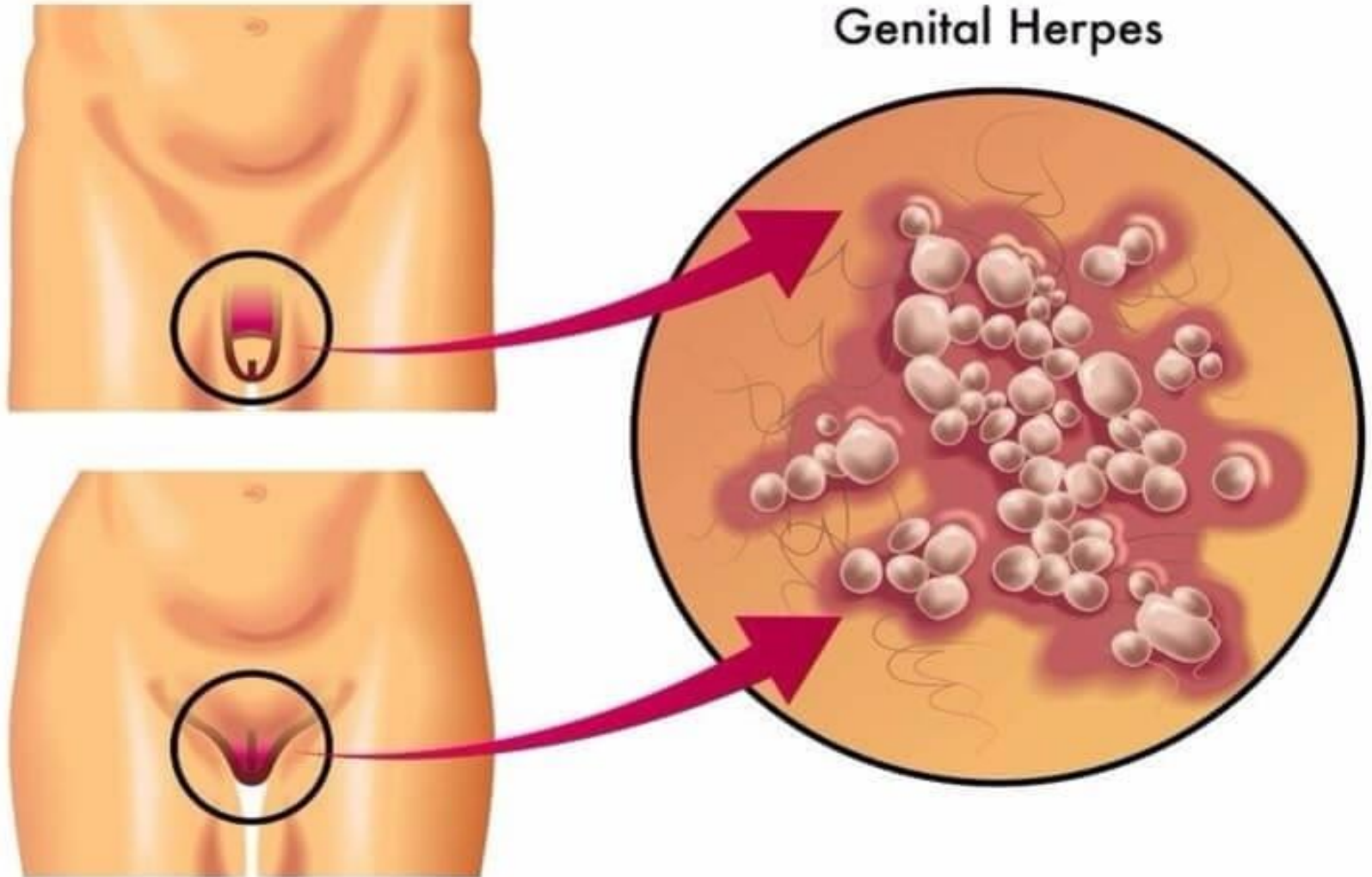
- A viral infection characterized by a localized primary lesion, latency and a tendency to localized recurrence.

Infectious agent

- Herpes simplex virus (HSV) type 2



HSV-2 Genital Herpes



Herpes Genitalia...Cont'd

Epidemiology

Occurrence

- worldwide.
- HSV 2 infection usually begins with sexual activity and is rare before adolescence, except in sexually abused children.
- Prevalence is greater (up to 60%) in lower socio-economic groups and persons with multiple sexual partners.

Reservoir – Humans.

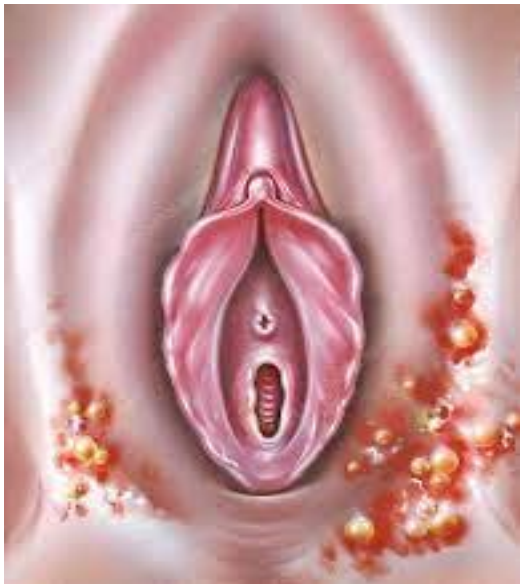
Herpes Genitalia...Cont'd

Mode of transmission

- Usually by sexual contact.
- Transmission to the neonate usually occurs via the infected birth canal, but less commonly occurs intrauterine or postpartum

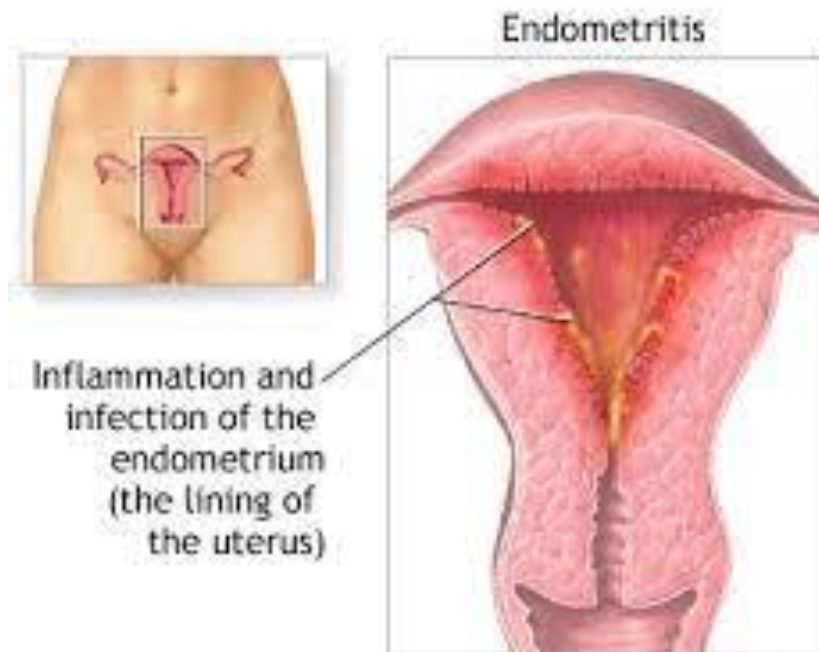
Clinical manifestation

- **First – episode** primary genital herpes is characterized by fever, headache, malaise and myalgias, pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal lymphadenopathy are the predominant local symptoms.
- Bilateral lesions of the external genitalia are characteristic: lesions may be vesicles, pustules, or painful erythematous ulcers.

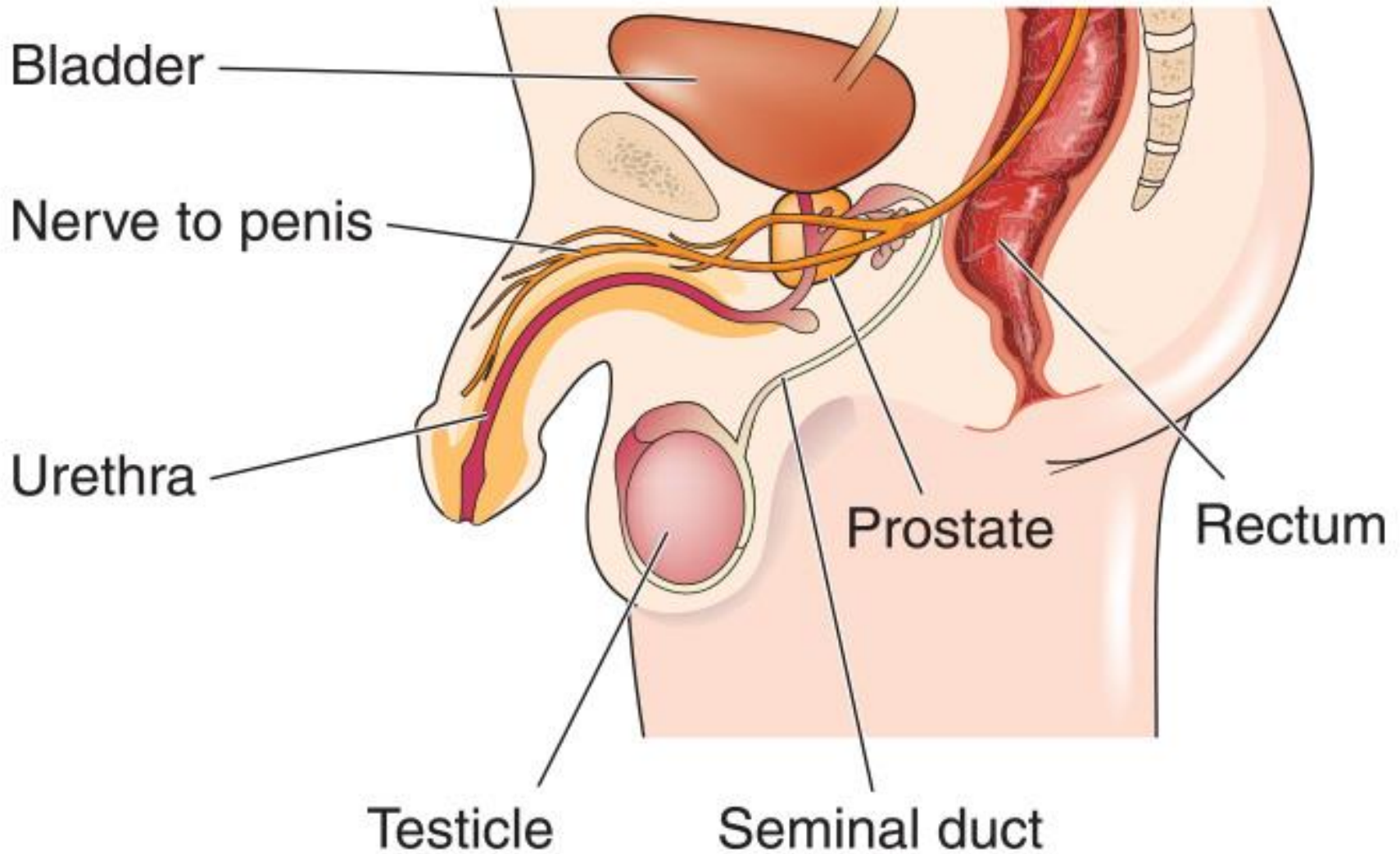


Clinical manifestation

- Cervix and urethra are involved in more than 80% of women with first episode infection.
- A clear mucoid discharge and dysuria are characteristics/symptoms of urethritis.
- Occasionally, HSV genital tract disease is manifested by endometritis and salpingitis in women and by prostatitis in men.



Clinical manifestation



Prevention and control

- Consistent use of condom is an effective means of reducing the risk of genital HSV– 2 transmission.
- Early diagnosis and treatment of cases

Gonorrhea

Definition

- An acute or chronic purulent infection of the urogenital tract.

Infectious agent

Neisseria gonorrhea, the gonococcus



Epidemiology

Occurrence

- Worldwide, affecting both genders, especially sexually active adolescents and young adults.
- Prevalent in communities of lower socio-economic status.

Reservoir - Strictly a human disease

Mode of transmission

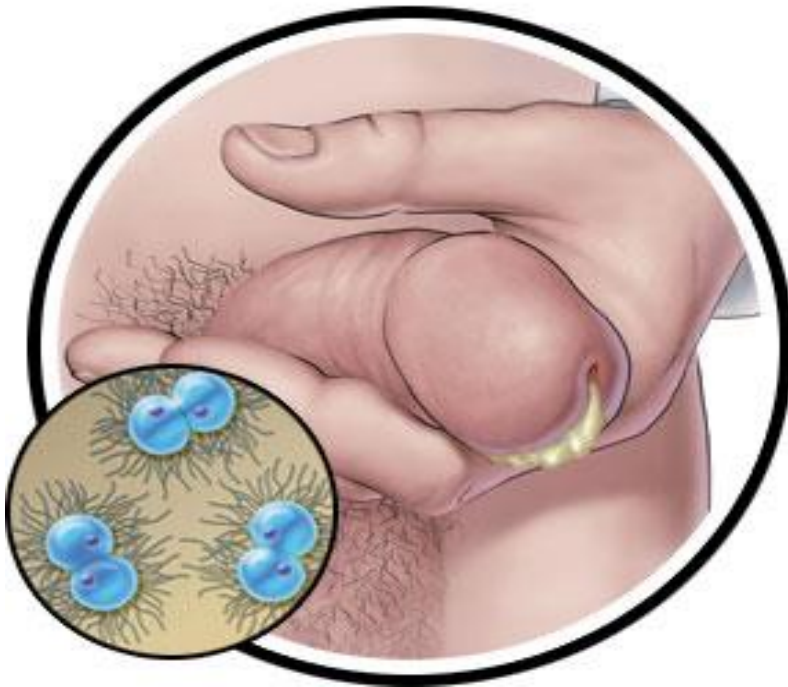
- Almost always as a result of sexual Activity

Gonorrhea...

Clinical manifestations

Males

- Usually involves the urethra resulting in purulent discharge, dysuria and frequency.



Females

- Females are usually asymptomatic.
- Vaginal discharge is common.
- Most common site of infection is cervix, followed by urethra, anal canal and pharynx.
- Neonates borne to infected mothers develop a purulent discharge which exudes from between eyelids which are edematous and erythematous 2 -3 days postpartum (**neonatorum ophthalmica**).



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Chapter 6:- STI

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Neonatorum Ophthalmica



Prevention and control

- Treatment of cases and contacts
- Health education on safe sex
- Controlling STDs among commercial sex workers
 - ✓ Monthly check up and treatment of cases
 - ✓ Provision of condom
- Screening of pregnant women and early treatment to prevent newborn infection
- Application of 1% tetracycline in both eyes of newborn as soon as delivered.

Syndromic Management of STI

Syndromic Management of STI

- There are two basic approaches in the management of STIs namely
 1. **Etiologic diagnosis** using laboratory tests and
 2. **Syndromic** approach.

Syndromic Management...cont'd

- The syndromic case management has the following key features:
 - It enables all trained first line health care providers to diagnose STI syndromes and treat patients on the spot, without waiting for laboratory results.
 - It will help to offer treatment on the initial visit which is an important step to stop the spread of the disease.
 - It is problem oriented (it responds to the patient's symptoms).
 - It is highly sensitive and does not miss mixed infections.
 - Uses flow charts that guide the health worker through logical steps.
 - Provides opportunity and time for education and counseling.

Syndromic Management...cont'd

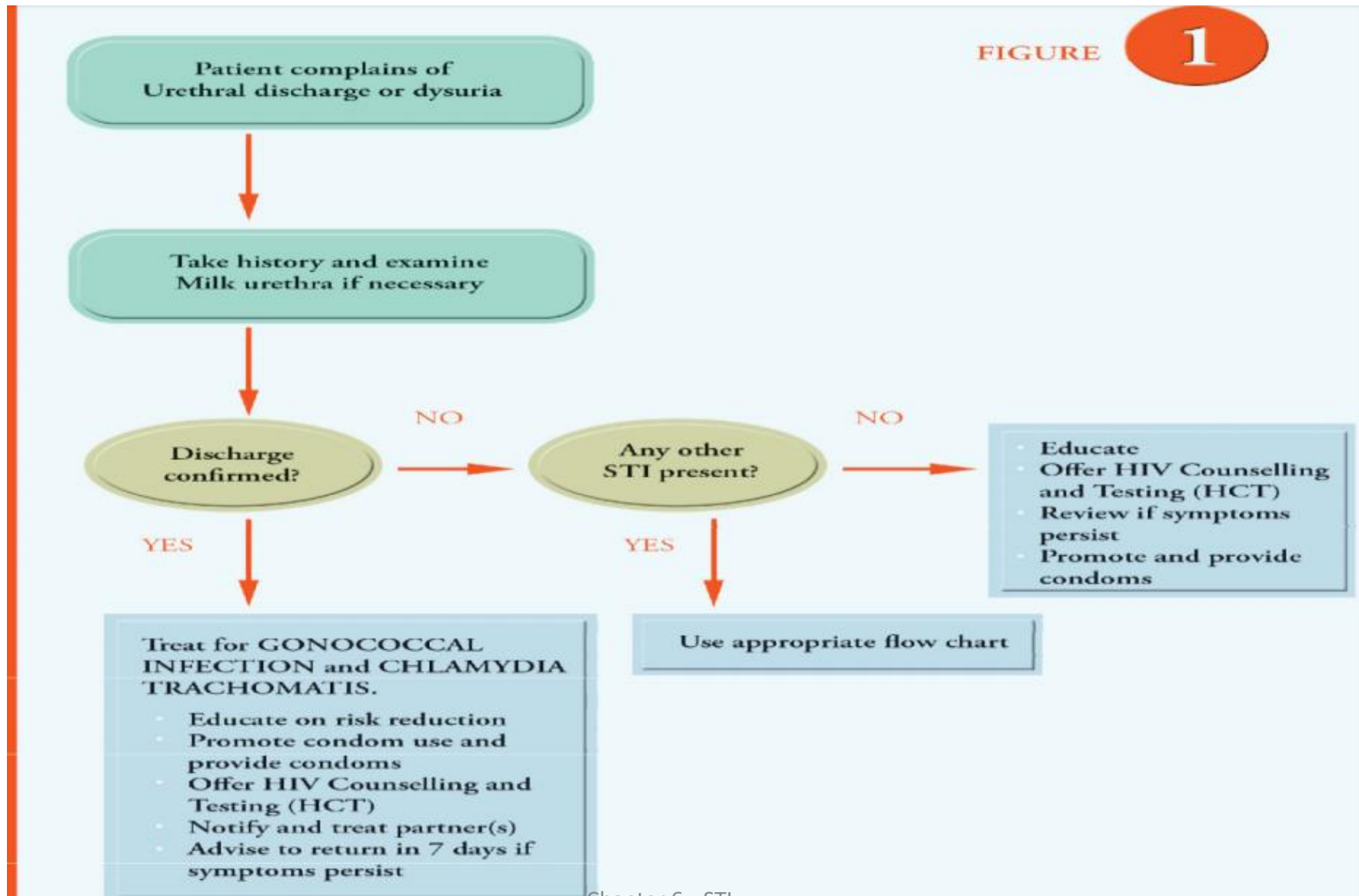
- The aim of syndromic STI management is to identify one of the **seven syndromes** and manage accordingly.
- **These are**
 1. vaginal discharges
 2. urethral discharges,
 3. genital ulcer,
 4. lower abdominal pain,
 5. scrotal swelling,
 6. inguinal bubo and
 7. neonatal conjunctivitis.

1. Urethral discharge syndrome

□ Cause or etiology of UDS:-

- The two most common causative agents of the syndrome are
 1. Neisseria gonorrhea and
 2. Chlamydia trachomatis

The algorithm of syndromic case management of urethral discharge syndrome (Adopted from the Ethiopian STI Guideline)



Treatment of UDS

❑ First line (preferred)

- Ceftriaxone 250mg IM stat **plus** Azithromycin 1gm po stat

❑ Alternative

- **Ciprofloxacin**, 500mg PO stat/**Spectinomycin** 2 gm IM stat

❖ PLUS

- **Doxycycline** 100 mg po bid for 7 days/**Tetracycline** 500 mg po QID for 7 days/**Erythromycin** 500 mg po QID for 7 days in cases of contraindications for Tetracycline (e.g. for children and pregnancy).

- ❑ **NB:** Patients should be advised to return if symptoms persist for 7 days after the initiation of treatment. Single dose treatment is encouraged as much as possible

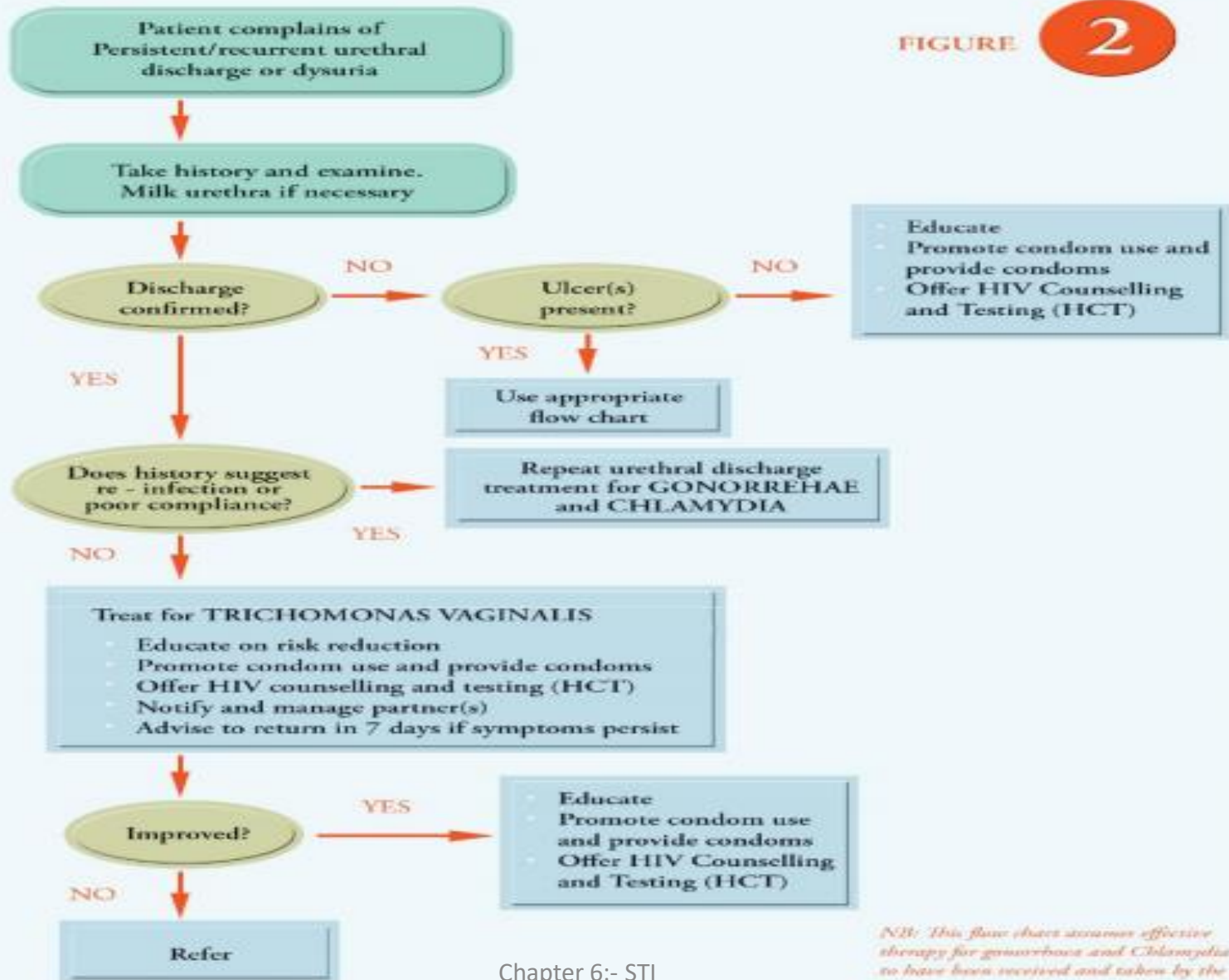
Persistent/Recurrent Urethral Discharge

- **Metronidazole** 2 gm po. stat/**Tinidazole** 1gm po once for 3 days (Avoid Alcohol!) **PLUS**
- **Azithromycin** 1 g orally in a single dose.

PERSISTENT/RECURRENT URETHRAL DISCHARGE IN MEN

FIGURE

2



NOTE: This flow chart assumes effective therapy for gonorrhoea and Chlamydia to have been received and taken by the patient prior to this consultation.

Genital Ulcer Syndrome (GUS)

- Genital ulcer is an open sore or a break in the continuity of the skin or mucous membrane of the genitalia as a result of sexually acquired infections.
- Genital ulcer facilitates transmission of HIV more than other sexually transmitted infections because it disrupts continuity of skins and mucous membranes significantly.

Genital Ulcer Syndrome...cont'd

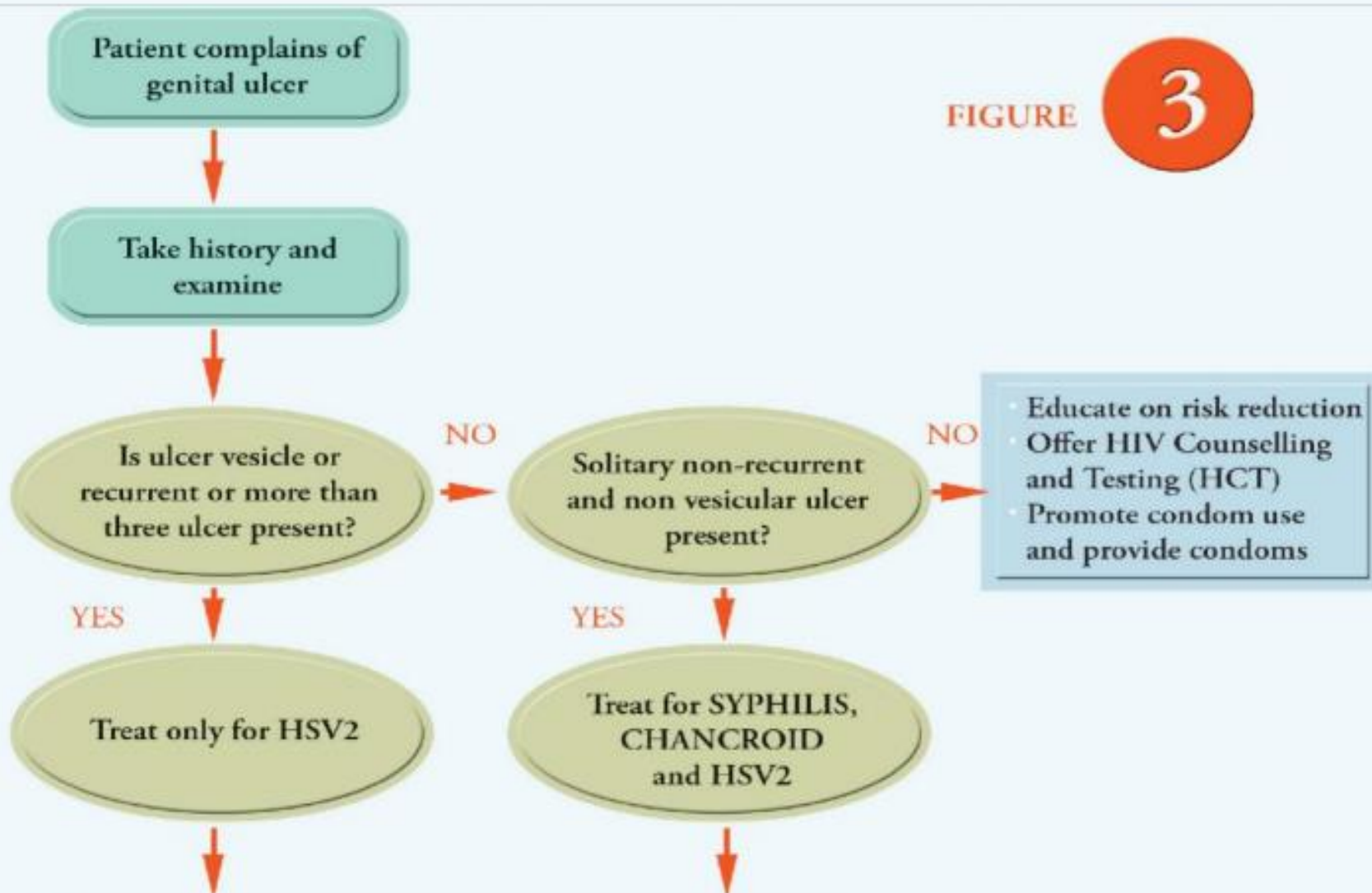
❑ Etiology of Genital Ulcer Syndrome

- There are different kinds of bacteria and viruses which commonly cause genital ulcer.
- Some of the common etiologies of genital ulcer syndrome are:
 - Herpes simplex virus (HSV-1 and HSV-2)
 - Treponema pallidum
 - Haemophilus ducreyi (Chancroid)
 - Chlamydia trachomatis serovar L1, L2 & L3 (LGV)
 - Klebsiella granulomatis

GENITAL ULCERS SYNDROME

FIGURE

3



-
- ```
graph TD; A[] --> B["• Educate on risk reduction
• Promote condom use and provide condoms
• Offer HIV Counselling and Testing (HCT)
• Ask the patient to return in 7 days
• Notify and manage partner(s)"]; B --> C{Ulcer(s) healed?}; C -- YES --> D["• Educate on risk reduction
• Promote condom use and provide condom
• Offer HIV Counselling and Testing (HCT)
• Notify and manage partner(s)"]; C -- NO --> E{Ulcers improving?}; E -- YES --> F["• Continue treatment to complete 10 days course.
• Consider followup"]; E -- NO --> G[Refer]; D --> F;
```
- Educate on risk reduction
  - Promote condom use and provide condoms
  - Offer HIV Counselling and Testing (HCT)
  - Ask the patient to return in 7 days
  - Notify and manage partner(s)

Ulcer(s) healed?

NO

Ulcers improving?

NO

Refer

YES

- Educate on risk reduction
- Promote condom use and provide condom
- Offer HIV Counselling and Testing (HCT)
- Notify and manage partner(s)

YES

- Continue treatment to complete 10 days course.
- Consider followup

# Pharmacologic Rx of Genital Ulcer

- The recommended treatment regimen for genital ulcer in Ethiopia is depicted below.

## ❑ **Treatment for non- vesicular genital ulcer**

- **Benzathine penicillin** 2.4 million units IM stat/**Doxycycline** (in penicillin allergy) 100mg bid for 14 days, PLUS
- **Ciprofloxacin** 500mg bid orally for 3 days /**Erythromycin** 500mg tab QID for 7 days PLUS
- **Acyclovir** 400mg TID orally for 10 days (or 200 mg five times per day of 10 day)

### ❑ Treatment for vesicular, multiple or recurrent genital ulcer

- Acyclovir 200 mg five times per day for 10 days Or Acyclovir 400 mg TID for 7 days
- **N.B.** There is no medically proven role for topical acyclovir, its use is discouraged.

# *Pharmacologic...cont'd*

## ❑ **Treatment for recurrent infection episodes:**

- Acyclovir 400 mg P.O. TID for 5 to 7 days,

## ❑ **Suppressive treatment:** recommended for patients with 6 recurrences or more per year

- Acyclovir, 400mg P.O. BID for 1 year
- **N.B.** The need for continued suppressive therapy should be reassessed.

# Vaginal Discharge

- Abnormal vaginal discharge in terms of quantity, color or odor could be most commonly as a result of vaginal infections.
- The most common causes of vaginal discharge are
  - ✓ *Neisseria gonorrhoeae*,
  - ✓ *Chlamydia trachomatis*,
  - ✓ *Trichomonas vaginalis*,
  - ✓ *Gardnerella vaginalis* (Polymicrobial),
  - ✓ *Candida albicans*.

# *Pharmacologic Vaginal Discharge*

- If **Risk assessment positive**:
  - ✓ **Ceftriaxone** 250mg IM stat/**Ciprofloxacin** 500mg po stat/ **Spectinomycin** 2 gm IM stat **PLUS**
  - ✓ **Azithromycin** 1gm po stat/**Doxycycline** 100 mg po bid for 7 days **PLUS**
  - ✓ **Metronidazole** 500 mg bid for 7 days
- If discharge is white or curd-like add **Clotrimazole** vaginal pessary 200 mg at bed time for 3 days
- **Note:** The preferred regimen is **Ceftriaxone** 250mg IM stat + **Azithromycin** 1gm po stat + **Metronidazole** 500 mg bid for 7days

# *Pharmacologic Vaginal Discharge*

## ❑ If risk assessment negative

- **Metronidazole** 500 mg bid for 7 days
- If discharge is white or curd-like add
- **Clotrimazole** vaginal pessary 200 mg at bed time for 3 days, OR
- **miconazole** vaginal pessary 200mg at bed time for 3 days.

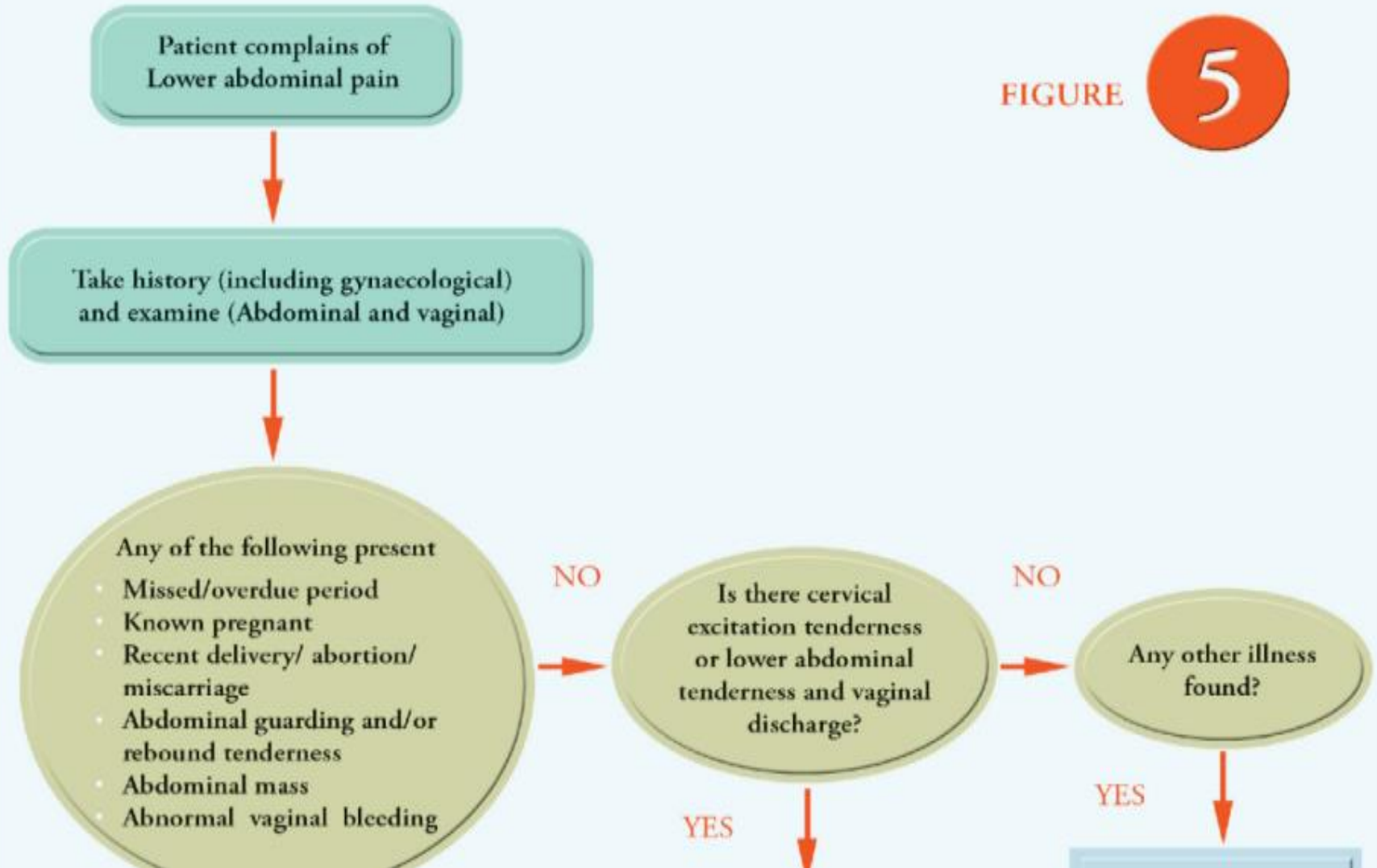
# Lower Abdominal Pain

- All sexually active women presenting with lower abdominal pain should be carefully evaluated for the presence of upper genital tract infections (tube, uterus, ovaries, and pelvic cavity).

# LOWER ABDOMINAL PAIN

FIGURE

5



YES

Refer patient for surgical or  
Gynaecological opinion and  
Assessment

Before referral set up an IV  
Line and resuscitate the  
patient if required.

Manage for PID  
Review in 3 days

Manage  
appropriately

NO

Patient has  
improved?

Admit or refer  
patient for  
admission

YES

Continue treatment until completed

- Educate on risk reduction
- Offer HIV Counselling and Testing (HCT)
- Promote condom use and provide condom
- Ask patient to return if necessary
- Notify and manage partner

# Treatment

## ❑ Pharmacologic For outpatients

- **Ceftriaxone** 250 mg IM stat/ Ciprofloxacin, 500mg po single dose/Spectinomycin 2gm i.m stat **PLUS**.
- **Azithromycin** 1gm po stat/Doxycycline 100 mg po B.I.D for 14 days **PLUS**
- **Metronidazole** 500 mg po b.i.d for 14 days
- Admit if there is no improvement within 72 hours

❑ **Note:** The preferred regimen is **Ceftriaxone** 250mg IM stat **PLUS**

❑ **Azithromycin** 1gm po stat **PLUS**

❑ **Metronidazole** 500 mg bid for 14 days

### ❑ For hospitalized patients (inpatients)

- **Ceftriaxone** 250 mg I.M/I.V /**Spectinomycin** 2 gm I.M bid PLUS
- **Azithromycin** 1gm po daily /**Doxycycline** 100 mg po b.i.d for 14 days PLUS
- **Metronidazole** 500 mg po b.i.d for 14 days
- **Note:** For inpatient PID, **ceftriaxone**, **spectinomycin** or **azithromycin** should continue for 24hrs after the patient remain clinically improved, after which **doxycycline** and **metronidazole** should continue for a total of 14 days

# Scrotal Swelling

- Inflammation of the epididymis usually manifests with acute onset of unilateral testicular swelling, often with tenderness of the epididymis and vas deferens, and occasionally with erythema and edema of the overlying skin.
- In older people the etiologic agent may be non-STIs such as
  - *E. coli*,
  - *Klebsiella* spp. or
  - *Pseudomonas*.
- TB orchitis is generally accompanied by an epididymitis

# Treatment

## ❑ Pharmacologic

### ❖ First line (preferred)

- Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat

### ❖ Alternative

- Ciprofloxacin, 500mg PO stat/Spectinomycin 2 gm IM stat PLUS
- Doxycycline 100 mg po bid for 7 days/Tetracycline 500 mg po QID for 7 days/Erythromycin 500 mg po QID for 7 days in cases of contraindications for Tetracycline (e.g. for children and pregnancy)

# Inguinal Bubo

- This is a painful, fluctuant, swelling of the lymph nodes in the inguinal region (groin).
- Buboes are usually caused by either chancroid or LGV.
- **Etiology**
- The common causes of inguinal and femoral bubo are:
  - *Chlamydia trachomatis* (L1, L2 and L3) (causes LGV)
  - *Klebsiella granulomatis* (donovanosis)
  - *Treponema pallidum* (causes syphilis)
  - *Haemophilus ducreyia* (causes chancroid)

# Treatment

## ❑ Pharmacologic

### ❑ First line

- **Ciprofloxacin**, 500mg P.O, BID for 3days PLUS
- **Doxycycline**, 100mg P.O, BID for 14days OR
- **Erythromycin**, 500mg, P.O, QID for 14days
- If patient have genital ulcer, add **Acyclovir** 400mg tid orally for 10 days (or 200mg five times per day for 10 days)
- **N.B.** Fluctuant lymph nodes should be aspirated through healthy skin but incision and drainage or excision of nodes may delay healing and should not be attempted.

# Individual assignment

- 1. HIV/AIDS**
- 2. Hepatitis B virus**

# **Thank you**